Synopsis

Machine Learning Based Loan Default Prediction System

Problem Id:02

Team member 1	Sidharth: Ph.No: 9416475283 Email:sidharthynr.sharma@gmail.com@gmail.com
Team member 2	Nishtha tyagi : Ph. No:6396847116 Email : nisthatyagii29@gmail.com
Team member 3	Piyush Shrivastava : Ph.No: 9709142559 Email : piyushsrivastava7143@gmail.com

- ## Problem Description:

Loan default prediction is an important challenge in the banking and financial industry. Banks give loans to customers after reviewing their financial background and credit history. However, not every customer is able to repay the loan on time, which can lead to financial losses for the bank.

The main aim of this project is to build a machine learning model that can predict whether a customer is likely to default on a loan. The prediction is made using different factors such as age, income, credit score, loan amount, employment status, loan term, and other financial details of the customer.

By analyzing past loan records, machine learning algorithms can identify patterns in customer behavior related to loan repayment. In this project, algorithms like Support Vector Machine (SVM) and K-Nearest Neighbors (KNN) are used to classify customers into two categories: those who are likely to repay the loan and those who may default.

This system can help banks and financial institutions make better decisions when approving loans and reduce the overall risk of loan defaults.

- ## Steps of Implementation:

1. Data Collection

The first step of the project involves collecting a dataset that contains information about customers and their loan details. The dataset is stored in a CSV file and includes various attributes such as financial background, loan information, and repayment status.

2. Data Preprocessing

Before applying machine learning algorithms, the dataset is cleaned and prepared for analysis. This includes handling missing values, converting categorical data into numerical form using encoding techniques, and scaling the numerical features using StandardScaler to ensure that all variables are on a similar scale.

3. Train-Test Split

The processed dataset is then divided into two parts: a training set and a testing set. The training data is used to train the machine learning models, while the testing data is used to evaluate how well the models perform on unseen data.

4. Dimensionality Reduction

To simplify the dataset and improve computational efficiency, Principal Component Analysis (PCA) is applied. PCA reduces the number of features while retaining the most important information from the original dataset.

5. Model Training

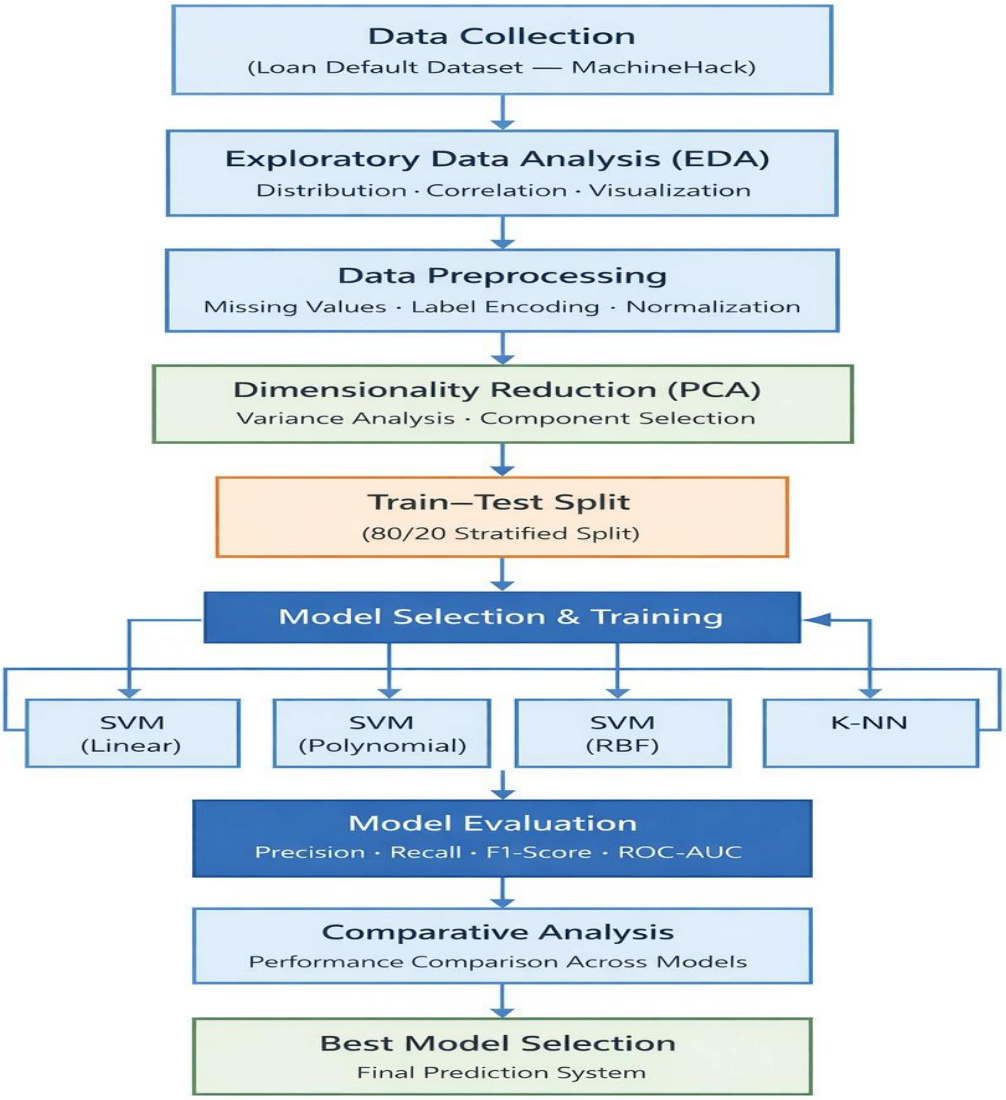
Machine learning models such as Support Vector Machine (SVM) and K-Nearest Neighbors (KNN) are trained using the training dataset. These models learn patterns from historical data to predict whether a customer is likely to default on a loan.

6. Model Evaluation

After training the models, their performance is evaluated using metrics such as accuracy, precision, recall, and F1-score. These metrics help measure how effectively the models can classify loan defaulters and non-defaulters.

7. Visualization and Analysis

Finally, the results are visualized using tools such as confusion matrices and graphical plots. These visualizations help in understanding the model's performance and comparing the results of different algorithms.



- ## Expected Modules/ Libraries to be used :

Pandas

Pandas is used to load the dataset from the CSV file and perform data manipulation tasks such as cleaning, organizing, and preparing the dataset for further analysis.

NumPy

NumPy is used for performing numerical operations and handling arrays efficiently during the data processing and machine learning stages.

Scikit-learn

Scikit-learn provides the machine learning tools used in this project. It is used to implement algorithms such as Support Vector Machine (SVM) and K-Nearest Neighbors (KNN), apply Principal Component Analysis (PCA), and evaluate the model using various performance metrics.

Matplotlib

Matplotlib is used to create graphical visualizations that help in understanding the results and performance of the machine learning models.

Seaborn

Seaborn is used for advanced visualizations such as heatmaps and confusion matrices, which make it easier to analyze and interpret the model's predictions.

- ## Expected Evaluation metrics and Plots to be used:

Accuracy

Accuracy represents the overall performance of the model. It shows how many predictions made by the model are correct compared to the total number of predictions.

Precision

Precision indicates how many of the predicted loan defaulters are actually correct. It helps measure the reliability of the model when it predicts that a customer will default on a loan.

Recall

Recall measures the model's ability to correctly identify customers who are actual loan defaulters. It is particularly important in financial applications where detecting risky customers is crucial.

F1 Score

The F1 Score is a balanced measure that combines both precision and recall. It provides a better understanding of model performance, especially when the dataset is imbalanced.

Confusion Matrix

A confusion matrix is used to visualize the performance of the classification model. It shows the number of correct and incorrect predictions for both classes, helping to understand how well the model distinguishes between defaulters and non-defaulters.

- **Visualization Plots**
- Confusion Matrix Heatmap
- Distribution Plots
- Correlation Heatmap

These visualizations help in analyzing the dataset and understanding the model's performance more clearly.

- ## Required topics from the subject:

Strings

Strings are used to handle textual or categorical information in the dataset, such as education level, employment type, and marital status. These values are processed and converted into numerical form so that they can be used by machine learning algorithms.

Files

Files are used to store and access the dataset required for the project. In this case, the loan dataset is stored in a CSV file, which is read into the program for analysis and model training.

Data Structures

Data structures such as DataFrames and arrays are used to organize and manage the dataset efficiently. They help in storing structured data and performing various data processing operations during the machine learning workflow.

(write 2 to 3 lines in each topic)

- ## Platform Required:

The project is implemented using the Python programming language. The development and experimentation are carried out using Jupyter Notebook, which provides an interactive environment for writing code, running machine learning models, and visualizing results step by step.

- ## Books and Link Sources:

Aurélien Géron – Hands-On Machine Learning with Scikit-Learn and TensorFlow

This book provides practical knowledge about machine learning techniques and how to implement them using Python libraries such as Scikit-learn.

Tom Mitchell – Machine Learning

A well-known reference book that explains the fundamental concepts and algorithms used in machine learning.

Scikit-learn Official Documentation-

This website provides detailed documentation and examples for implementing machine learning algorithms using the Scikit-learn library.

https://scikit-learn.org

Kaggle-

Kaggle is used as a resource for datasets, tutorials, and learning materials related to machine learning and data science.

https://www.kaggle.com

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